Mathematical Computations in Python - With and Without NumPys

Without NumPys

Python Computation - Without NumPys

In [1]:
```
arr = [10, 20, 30, 40, 50, 60, 70, 80, 90, 100]

print("Original Array:", arr)
```

```
Original Array: [10, 20, 30, 40, 50, 60, 70, 80, 90, 100]
```

Sum

In [2]:
```
total = 0

for num in arr:
    total += num

print("Sum:", total)
```

```
Sum: 550
```

Maximum Value

In [3]:
```
maximum = arr[0]

for num in arr:
    if num > maximum:
        maximum = num

print("Maximum:", maximum)
```

```
Maximum: 100
```

Minimum Value

In [4]:
```
minimum = arr[0]

for num in arr:
    if num < minimum:
        minimum = num

print("Minimum:", minimum)
```

```
Minimum: 10
```

Average

In [5]:
```
total = 0
count = 0

for num in arr:
    total += num
    count += 1

average = total / count

print("Average:", average)
```

```
Average: 55.0
```

Multiplying by 2

In [6]:
```
result = []

for num in arr:
    result.append(num * 2)

print("Result:", result)
```

```
Result: [20, 40, 60, 80, 100, 120, 140, 160, 180, 200]
```

Squaring Numbers

In [7]:
```
squares = []

for num in arr:
    squares.append(num ** 2)

print("Squares:", squares)
```

```
Squares: [100, 400, 900, 1600, 2500, 3600, 4900, 6400, 8100, 10000]
```

Standard Deviation

In [8]:
```
mean = sum(arr) / len(arr)

squared_diff = 0

for num in arr:
    squared_diff += (num - mean) ** 2

variance = squared_diff / len(arr)
std_dev = variance ** 0.5

print("Standard Deviation:", std_dev)
```

```
Standard Deviation: 28.722813232690143
```

Filtering Numbers based on Condition

In [9]:
```
filtered = []

for num in arr:
    if num > 50:
        filtered.append(num)

print("Filtered:", filtered)
```

```
Filtered: [60, 70, 80, 90, 100]
```

With NumPys

